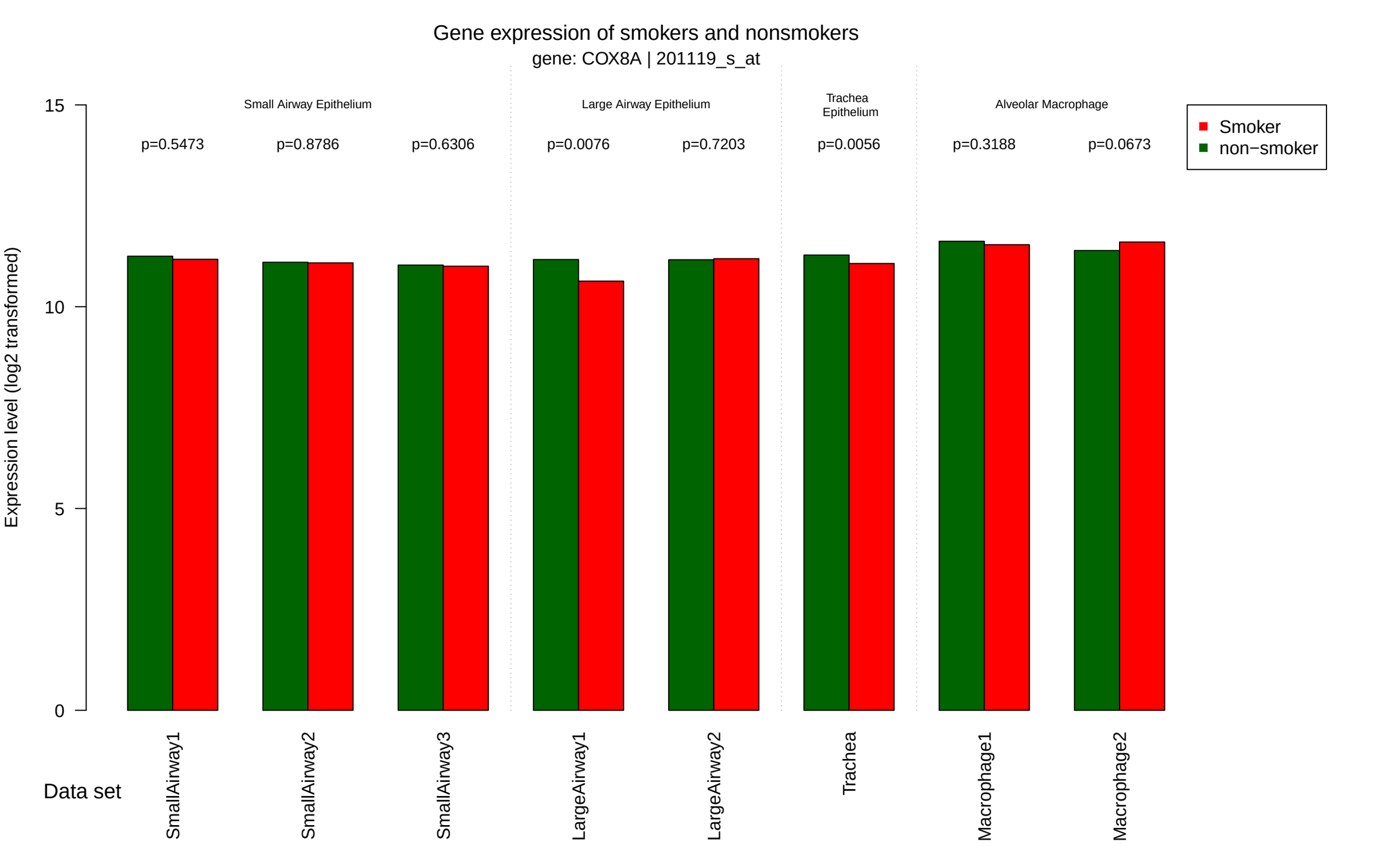
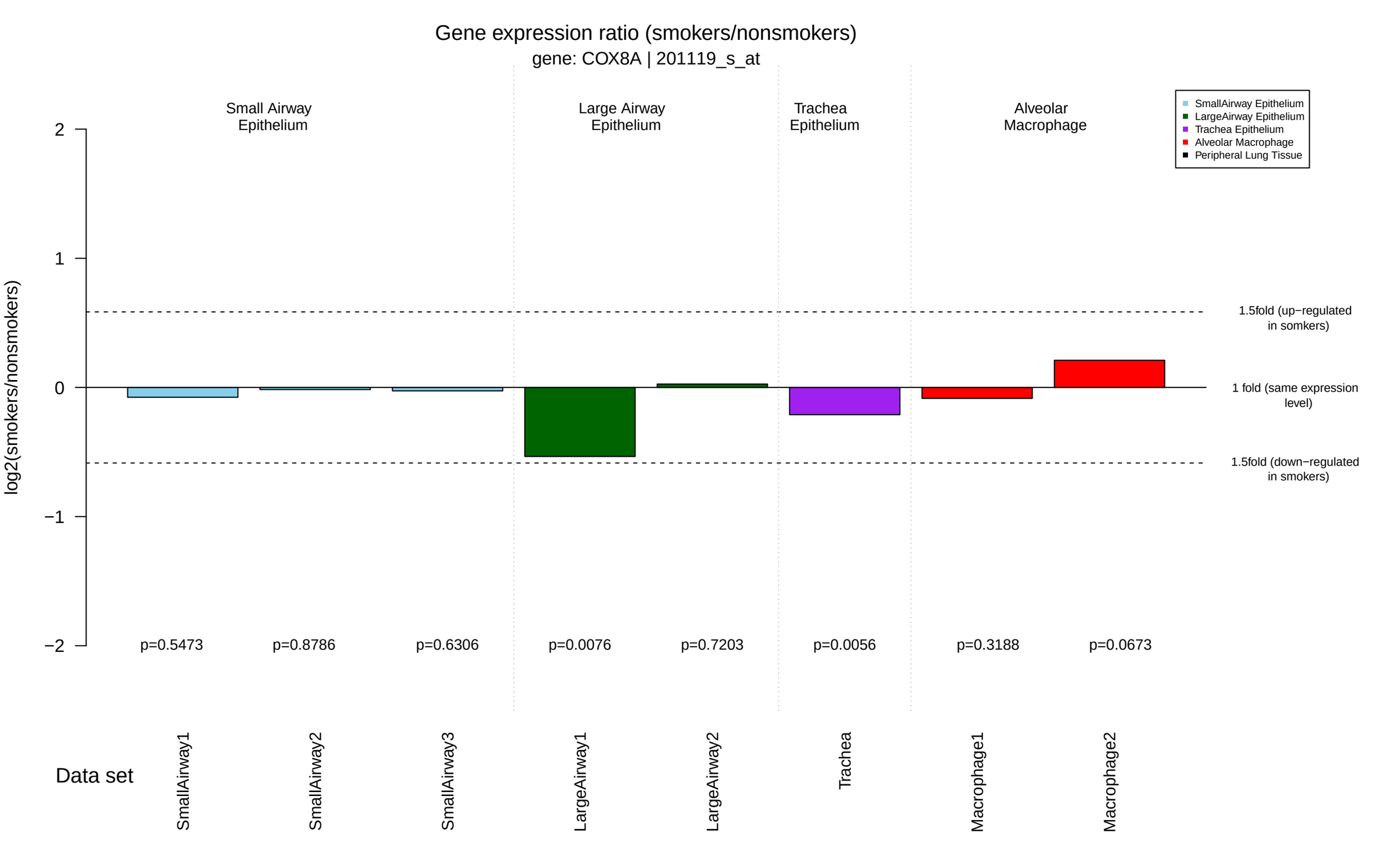
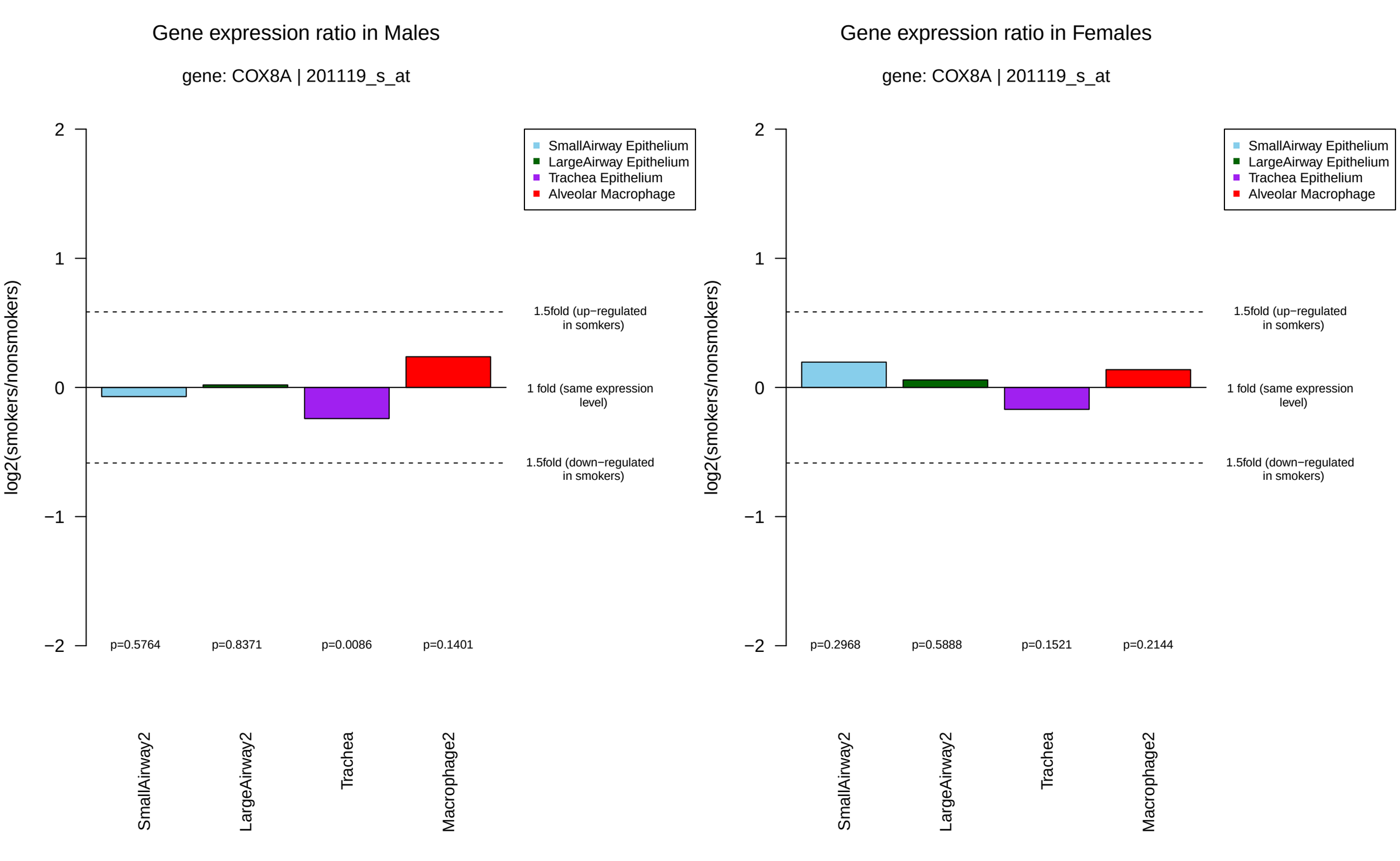
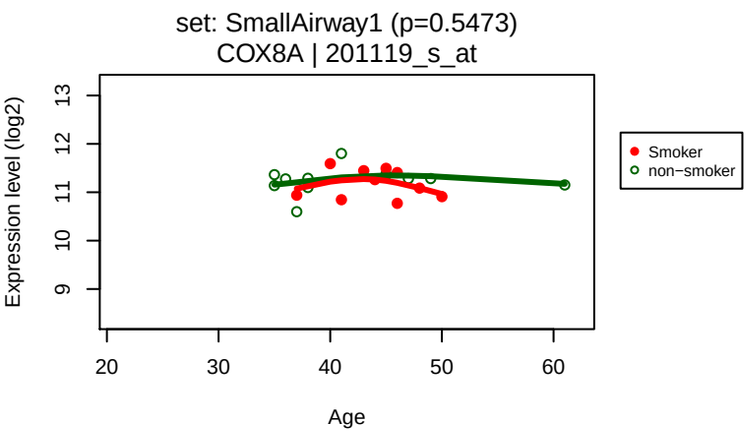
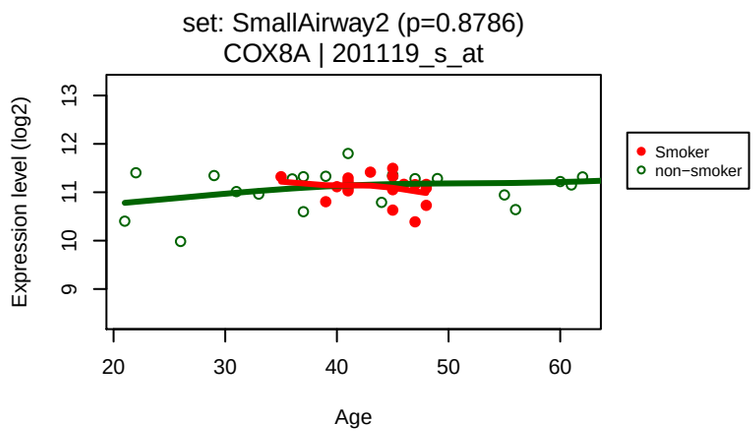
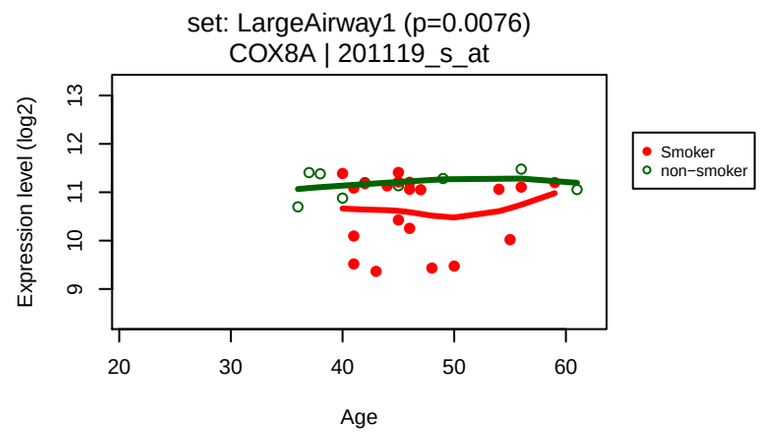

Gene expression of smokers and nonsmokers
gene: COX8A | 201119_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5473
p=0.8786
p=0.6306
p=0.0076
p=0.7203
p=0.0056
p=0.3188
p=0.0673
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

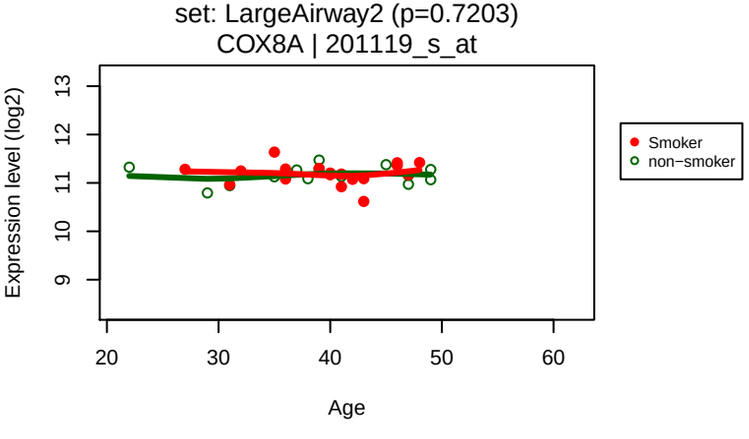

Gene expression ratio (smokers/nonsmokers)
gene: COX8A | 201119_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5473
p=0.8786
p=0.6306
p=0.0076
p=0.7203
p=0.0056
p=0.3188
p=0.0673
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

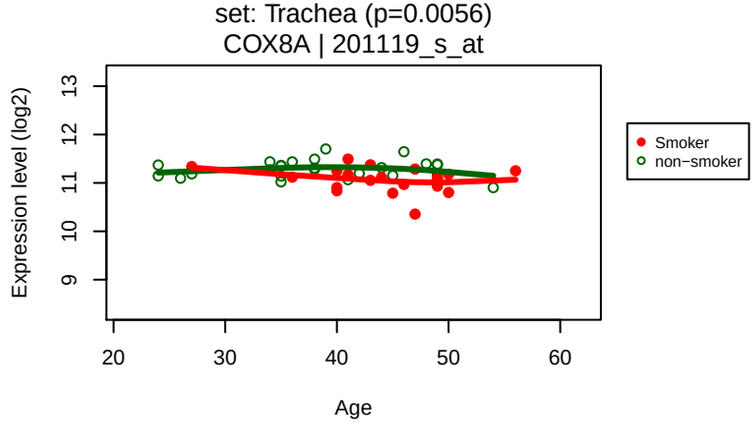

Gene expression ratio in Males
gene: COX8A | 201119_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5764
p=0.8371
p=0.0086
p=0.1401
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX8A | 201119_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2968
p=0.5888
p=0.1521
p=0.2144
SmallAirway2
LargeAirway2
Trachea
Macrophage2

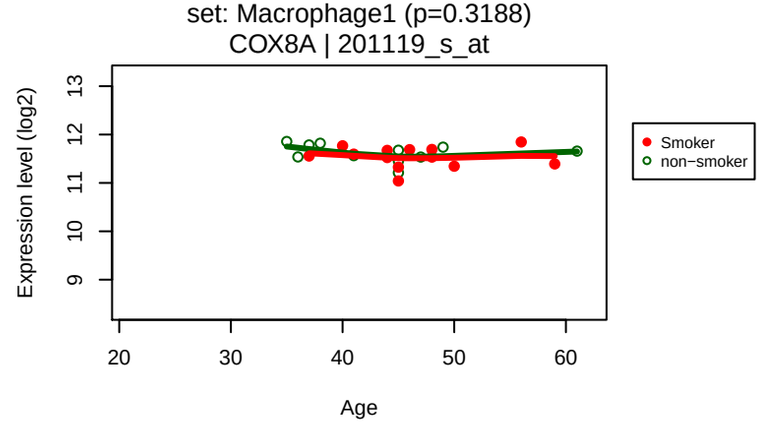

set: SmallAirway1 (p=0.5473)
COX8A | 201119_s_at
Expression level (log2)
Age
Smoker
non−smoker

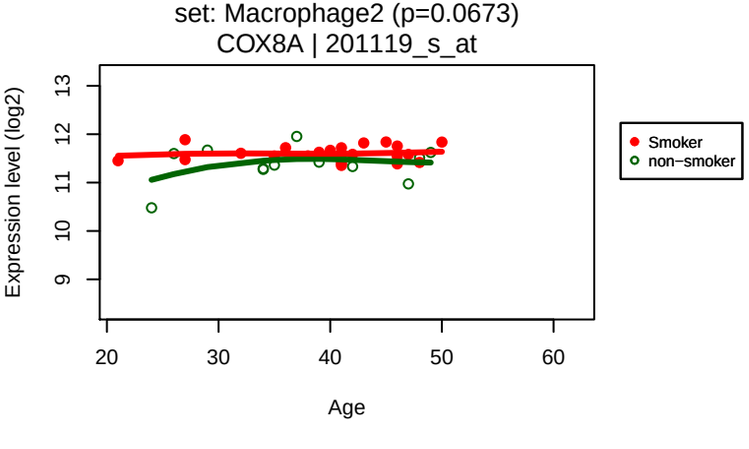

set: SmallAirway2 (p=0.8786)
COX8A | 201119_s_at
Expression level (log2)
Age
Smoker
non−smoker

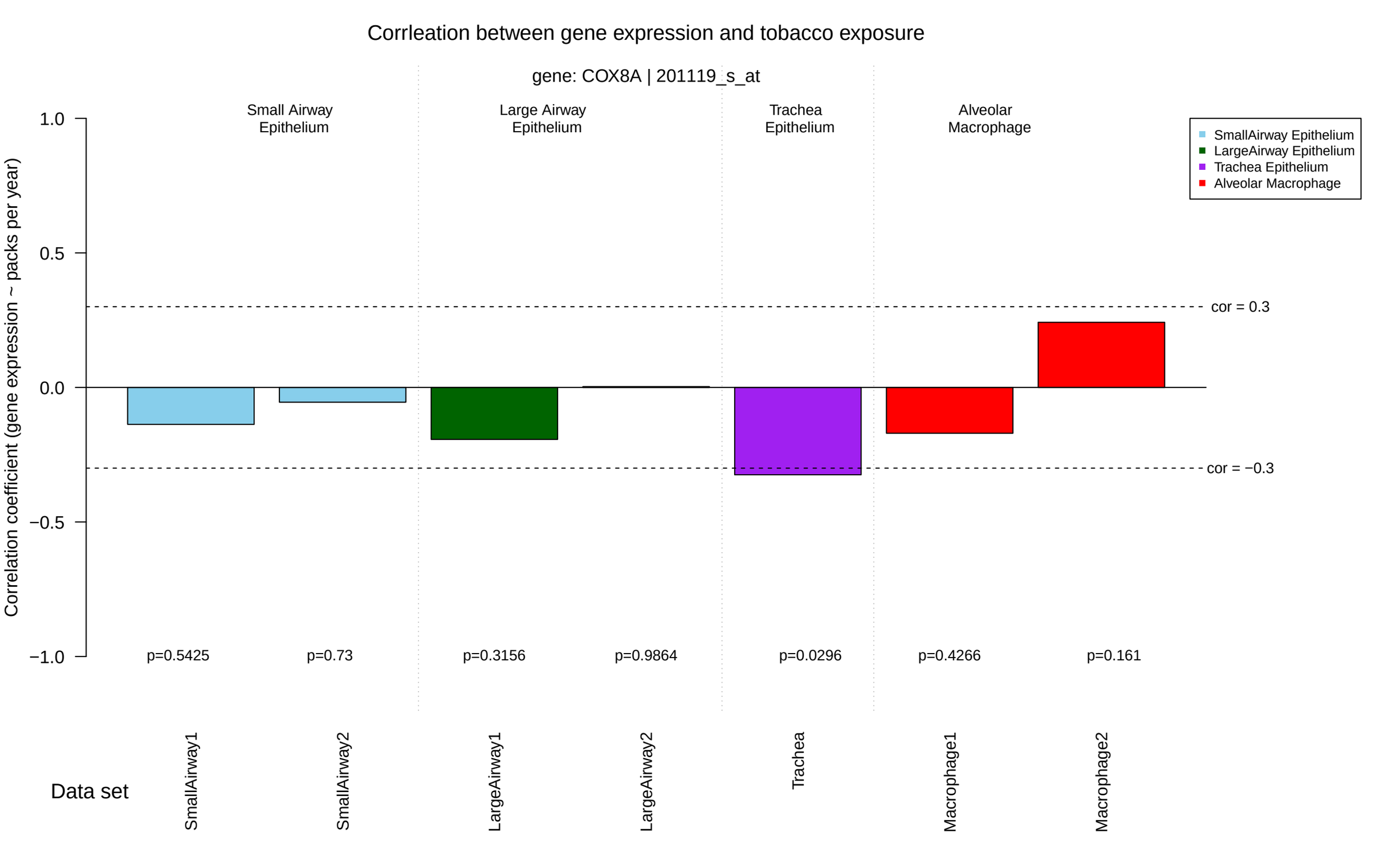

set: LargeAirway1 (p=0.0076)
COX8A | 201119_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7203)
COX8A | 201119_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0056)
COX8A | 201119_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3188)
COX8A | 201119_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0673)
COX8A | 201119_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX8A | 201119_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5425
p=0.73
p=0.3156
p=0.9864
p=0.0296
p=0.4266
p=0.161
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2